

BEE 4750 Lab 2: Monte Carlo

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Due Date

Thursday, 10/02/25, 9:00pm

Setup

The following code should go at the top of most Julia scripts; it will load the local package environment and install any needed packages. You will see this often and shouldn't need to touch it.

```
import Pkg
Pkg.activate(".")
Pkg.instantiate()
```

Activating project at `~/Lab2 Copy/lab-2-pdc76`

```
using Random # random number generation
using Distributions # probability distributions and interface
using Statistics # basic statistical functions, including mean
using Plots # plotting
```

Overview

In this lab, we will conduct a Monte Carlo experiment and explore how sensitive Monte Carlo estimates are to the underlying probability distribution(s).

You should always start any computing with random numbers by setting a “seed,” which controls the sequence of numbers which are generated (since these are not *really* random, just “pseudorandom”). In Julia, we do this with the `Random.seed!()` function.

```
Random.seed!(1)
```

`TaskLocalRNG()`

It doesn't matter what seed you set, though different seeds might result in slightly different values. But setting a seed means every time your notebook is run, the answer will be the same.

Seeds and Reproducing Solutions

If you don't re-run your code in the same order or if you re-run the same cell repeatedly, you will not get the same solution. If you're working on a specific problem, you might want to re-use `Random.seed()` near any block of code you want to re-evaluate repeatedly.

Probability Distributions and Julia

Julia provides a common interface for probability distributions with the [Distributions.jl package](#). The basic workflow for sampling from a distribution is:

1. Set up the distribution. The specific syntax depends on the distribution and what parameters are required, but the general call is the similar. For a normal distribution or a uniform distribution, the syntax is

```
# you don't have to name this "normal_distribution"
#  is the mean and  is the standard deviation
normal_distribution = Normal( , )
# a is the upper bound and b is the lower bound; these can be set to +Inf or -Inf for an
uniform_distribution = Uniform(a, b)
```

There are lots of both [univariate](#) and [multivariate](#) distributions, as well as the ability to create your own, but we won't do anything too exotic here.

2. Draw samples. This uses the `rand()` command (which, when used without a distribution, just samples uniformly from the interval $[0, 1]$.) For example, to sample from our normal distribution above:

```
# draw n samples
rand(normal_distribution, n)
```

Putting this together, let's say that we wanted to simulate 100 six-sided dice rolls. We could use a [Discrete Uniform distribution](#).

```
dice_dist = DiscreteUniform(1, 6) # can generate any integer between 1 and 6
dice_rolls = rand(dice_dist, 100) # simulate rolls
```

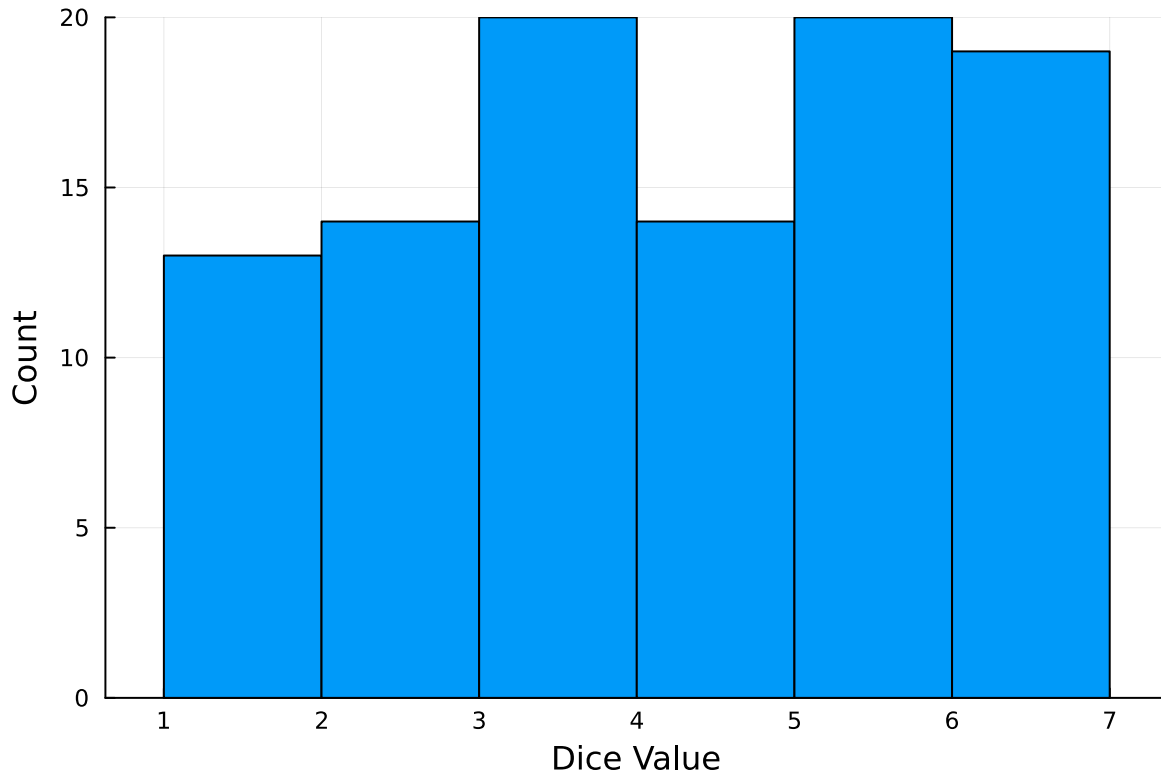
100-element Vector{Int64}:

```
1
3
5
4
6
2
5
5
5
2

3
6
5
5
6
3
6
6
6
```

And then we can plot a histogram of these rolls:

```
histogram(dice_rolls, legend=:false, bins=6)
ylabel!("Count")
xlabel!("Dice Value")
```



Instructions

Remember to:

- Evaluate all of your code cells, in order (using a `Run All` command). This will make sure all output is visible and that the code cells were evaluated in the correct order.
- Tag each of the problems when you submit to Gradescope; a 10% penalty will be deducted if this is not done.

Problem

A common engineering problem is to quantify flood risk, which is typically computed by propagating a flood hazard distribution through a *depth-damage function* relating flood depths to economic damages. A reasonable depth-damage function for a house without a basement is a bounded logistic function,

$$d(h) = \mathbb{1}_{h>0} \frac{L}{1 + \exp(-k(h - h_0))},$$

where d is the damage as a percent of total structure value, h is the water depth (relative to the house's ground floor elevation) in m, $\mathbb{1}_{x>0}$ is the indicator function, L is the maximum loss in USD, k is the slope of the depth-damage relationship, and h_0 is the inflection point. We'll assume $L = \$200,000$, $k = 0.8$, and $h_0 = 3$.

For this problem, suppose that we have two different probability distributions characterizing annual maxima flood depths:

1. $h \sim \text{LogNormal}(1.2, 0.3)$;
2. $h \sim \text{GeneralizedExtremeValue}(3, 0.9, -0.15)$.

Plot histograms of samples from these distributions and conduct a Monte Carlo experiment for annual maximum flood damages using each. Answer the following questions:

- What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?
- How did you decide on your sample size?
- Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)?
- How might you decide (based on getting additional information, if needed) which distribution is “better”?

```
#Initialization
L = 200000 #max loss from damage in USD
k = 0.8 #Slope-depth parameter
ho = 3 #Inflection point

#Damage function
function damage(h)
    if h < 0
        d = 0
    else
        d = 1 * L / (1 + exp(-k * (h - ho)))
    end
    return d
end
end
```

damage (generic function with 1 method)

Above is the function for damage as laid out in the problem setup. It includes the initialized variables, L, K, h_0 . It returns zero for $h < 0$, and the damage function evaluated at a height, h , otherwise.

```
log_normal = LogNormal(1.2, 0.3) #log-normal distribution with log mean 1.2 and stddev 0.3
generalized_ev = GeneralizedExtremeValue(3, 0.9, -0.15) #generalized extreme value distribut
```

```
GeneralizedExtremeValue{Float64}(=3.0, =0.9, =-0.15)
```

Creates variables that call the probability distributions, LogNormal and GeneralizedExtremeValue.

```
#Simulate 100000 heights for each of the distributions
log_normal_heights = rand(log_normal, 100000)
log_damages = damage.(log_normal_heights)

generalized_ev_heights = rand(generalized_ev, 100000)
gev_damages = damage.(generalized_ev_heights)

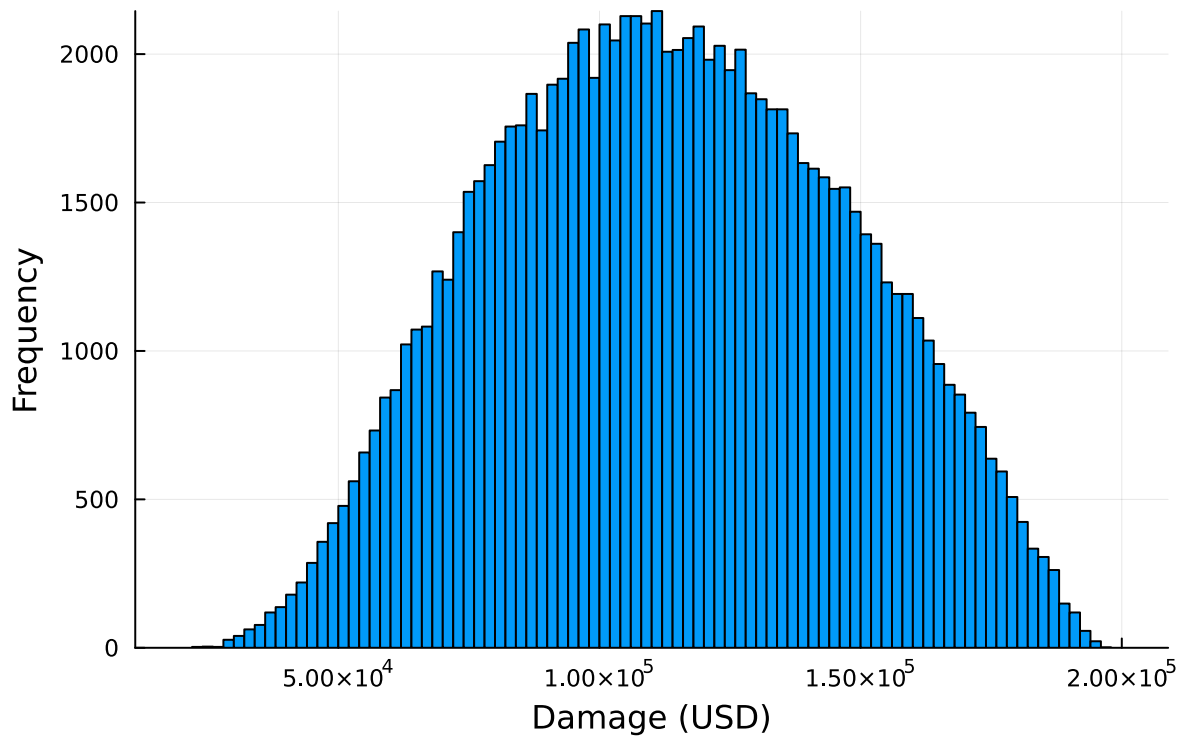
# LogNormal damages histogram
log_graph = histogram(
    log_damages;
    xlabel="Damage (USD)",
    ylabel="Frequency",
    title="Histogram of Flood Damages w/ LogNormal Depths",
    legend=false)

# GEV damages histogram
ev_graph = histogram(
    gev_damages;
    xlabel="Damage (USD)",
    ylabel="Frequency",
    title="Histogram of Flood Damages w/ GEV Depths",
    legend=false)

display(log_graph)
display(ev_graph)
```

A histogram showing the frequency distribution of damage in USD. The x-axis is labeled 'Damage (USD)' and ranges from 0 to 2.00 × 10⁵ with major ticks at 5.00 × 10⁴, 1.00 × 10⁵, 1.50 × 10⁵, and 2.00 × 10⁵. The y-axis is labeled 'Frequency' and ranges from 0 to 5000 with major ticks at 0, 1000, 2000, 3000, 4000, and 5000. The histogram consists of 25 blue bars with black outlines. The distribution is unimodal and right-skewed, with a peak frequency of approximately 5200 occurring at a damage value of about 1.05 × 10⁵ USD. The frequency decreases as the damage value increases beyond the peak, with a long tail extending towards 2.00 × 10⁵ USD.

Histogram of Flood Damages w/ GEV Depths



Generate the histograms for LogNormal and GeneralizedExtremeValues with frequency on the vertical axis and damage cost on the horizontal axis.

```
# Calculate the statistics for mean and 99th quantile
#Log statistics
log_mean = mean(log_damages)
log_quant = quantile(log_damages, 0.99)

#GEV statistics
gev_mean = mean(gev_damages)
gev_quant = quantile(gev_damages, 0.99)

#Print the statistics for both distributions
println("LogNormal Mean Damage: \${$(round(log_mean, digits=2))}")
println("LogNormal 99th Percentile Damage: \${$(round(log_quant, digits=2))}")
println("GeneralizedExtremeValue Mean Damage: \${$(round(gev_mean, digits=2))}")
println("GeneralizedExtremeValue 99th Percentile Damage: \${$(round(gev_quant, digits=2))}")
```

LogNormal Mean Damage: \$115070.72

LogNormal 99th Percentile Damage: \$189728.66

GeneralizedExtremeValue Mean Damage: \$113566.46
GeneralizedExtremeValue 99th Percentile Damage: \$183424.03

```
N = 200000
sample_sizes = 0:1000:N
log_means = [mean(log_damages[1:n]) for n in sample_sizes]
gev_means = [mean(gev_damages[1:n]) for n in sample_sizes]
mean_graph = plot(
    sample_sizes,
    [log_means gev_means];
    xlabel="Number of Simulations",
    ylabel="Mean Damage (USD)",
    title="Mean Damage vs Number of Simulations",
    label=["LogNormal" "GEV"],
    legend=:topright)
display(mean_graph)
```

BoundsError: BoundsError: attempt to access 100000-element Vector{Float64} at index [1:101000]
BoundsError: attempt to access 100000-element Vector{Float64} at index [1:101000]

Stacktrace:

```
[1] throw_boundserror(A::Vector{Float64}, I::Tuple{UnitRange{Int64}})
      @ Base ./essentials.jl:14

[2] checkbounds
      @ ./abstractarray.jl:699 [inlined]

[3] getindex(A::Vector{Float64}, I::UnitRange{Int64})
      @ Base ./array.jl:936

[4] (::var"#15#16")(n::Int64)
      @ Main ./none:0

[5] iterate
```

```

@ ./generator.jl:48 [inlined]

[6] collect_to!(dest::Vector{Float64}, itr::Base.Generator{StepRange{Int64, Int64}, var"#15#16"}
    @ Base ./array.jl:849

[7] collect_to_with_first!(dest::Vector{Float64}, v1::Float64, itr::Base.Generator{StepRange{Int64, Int64}, var"#15#16"}
    @ Base ./array.jl:827

[8] collect(itr::Base.Generator{StepRange{Int64, Int64}, var"#15#16"})
    @ Base ./array.jl:801

[9] top-level scope
    @ ~/Lab2 Copy/lab-2-pdc76/jl_notebook_cell_df34fa98e69747e1a8f8a730347b8e2f_X26sZmlsZQ==.jl:1

```

References

Put any consulted sources here, including classmates you worked with/who helped you.

<https://juliastats.org/Distributions.jl/v0.14/univariate.html> - The documentation for Distributions.jl to help me understand the LogNormal and GeneralizedExtremeValue distributions.